

Problem Statement

Expanded research and problem analysis for Slide 1. The original slide establishes SIH 2026 Problem ID 26009 and the Space Technology / Software framing. [\[filecite\]turn1file0\[L2-L11\]](#)

1. What the problem actually asks

Smart India Hackathon 2026 Problem Statement 26009 is framed around using AI/ML and space technology to identify manganese reserves and overcome production shortfalls. The two halves of the problem should be treated as one decision chain: better knowledge of the ore body must translate into better production planning.

The problem is therefore broader than mineral detection. It includes exploration targeting, reserve estimation, operational forecasting, disruption detection and corrective planning.

2. Scale of the underlying mining system

MOIL’s FY2024–25 Annual Report reports 121.97 million tonnes of total reserves and resources as of 31 March 2025, comprising 53.47 Mt reserves and 68.50 Mt resources. It also reports 107,530 metres of exploratory drilling in FY2024–25, which added 16.07 Mt of resources. [\[cite\]turn0search48](#)

The same report identifies exploration as a foundation for future growth and gives a strategic production target of 3.50 million tonnes by 2030. This provides a measurable context for a system that aims to improve exploration targeting and production planning.

3. Why reserve identification is difficult

Ore is a three-dimensional subsurface phenomenon while satellite observations are predominantly surface or near-surface signals. Therefore NAKSHATRA-X should never claim that a satellite image directly proves a deep manganese reserve.

The correct scientific framing is multi-source evidence fusion: satellite-derived surface/environmental indicators + borehole/core information + geological maps + assays + historical mine information. Spatial modelling can then estimate grade or prospectivity while reporting uncertainty.

4. Why production shortfalls are a separate problem

Even if the geological model is correct, production can fall below plan because of weather, water accumulation, equipment availability, blasting delays, haulage constraints, grade constraints or other operational conditions.

NAKSHATRA-X therefore adds a temporal decision layer: detect changing conditions, estimate the likely production impact, simulate alternatives and recommend an operational response.

5. National relevance

MOIL’s FY25 MD&A; reports Indian manganese ore production of 3.75 million tonnes, up from 3.38 Mt in FY24, while manganese ore imports were about 6.58 million tonnes in FY25 compared with 5.59 Mt in FY24. [\[cite\]turn1search24](#)

IBM’s Indian Minerals Yearbook identifies MOIL among principal manganese-ore producers and lists operating locations including Balaghat in Madhya Pradesh and Bhandara/Nagpur in Maharashtra. [\[cite\]turn1search26](#)

6. NAKSHATRA-X problem formulation

Input: heterogeneous geological, satellite, weather and mine-operation data.

Inference: estimate reserve/grade/prospectivity and quantify uncertainty.

Prediction: forecast production risk and disruption probability.

Optimization: determine feasible blending or operational alternatives.

Action: deliver an explainable recommendation to the responsible mine planner/engineer.

7. Judge-ready problem statement

“Mining data exists across geology, remote sensing, weather and operations, but the decision-maker needs one evidence layer that connects them. NAKSHATRA-X converts these disconnected signals into uncertainty-aware

reserve intelligence and production-risk decisions.”

Evidence and exact source paths

ID	Source	Exact resource / URL	Use in NAKSHATRA-X
S1	MOIL 63rd Annual Report 2024-25	https://moil.nic.in/userfiles/Annual_Report_2024_25.pdf	Reserves/resources, exploration drilling, production and strategic target.
S2	MOIL MD&A; / Annual Report copy	https://moil.nic.in/userfiles/file/InvRel/Financials/Annual_Report_2024-25.pdf	FY25 manganese production and import context.
S3	MOSDAC — GSMap ISRO Rain	https://www.mosdac.gov.in/gsmmap-isro-rain	Exact rainfall product; hourly, 0.1° and March 2000 onward.
S4	GSI Reports database	https://gsi.gov.in/gsiReport/	Official geological/exploration report search.
S5	GSI Mineral Hunt data portal	https://hackathon.gsi.gov.in/login	Direct geology, exploration and remote-sensing data routes.
S6	GSI geology data	https://nis.bisag-n.gov.in/download/Geology/	Direct geology data directory.
S7	GSI exploration reports/data	https://nis.bisag-n.gov.in/download/Exploration_data_reports/	Direct exploration data/report directory.
S8	GSI remote sensing ASTER	https://nis.bisag-n.gov.in/download/Remotesensing_ASTER/	Direct remote-sensing directory.
S9	GSI/OGD manganese deposit dataset	https://tn.data.gov.in/catalog/location-manganese-ore-deposits-in-dia-and-its-salient-features	Specific manganese-deposit location/geology dataset.
S10	Indian Minerals Yearbook portal	https://ibm.gov.in/IBMPortal/pages/Indian_Minerals_Yearbook	Official IBM source for manganese statistics.
S11	Indian Minerals Yearbook 2024	https://ibm.gov.in/writereaddata/files/177426215469c1178a48453IMYB_2024_EBookFinal.pdf	Producer/location and manganese industry data.
S12	ESA — Sentinel-1 seeing through clouds	https://www.esa.int/ESA_Multimedia/Videos/2014/03/Sentinel-1_seeing_through_clouds	SAR capability through cloud/rain and day/night.
S13	ESA — Introducing Sentinel-1	https://www.esa.int/Applications/Observing_the_Earth/Copernicus/Sentinel-1/Introducing_Sentinel-1	C-band SAR, all-weather/day-night mission context.

Recommended evidence to add to your final appendix

Attach one sample borehole/assay record, one satellite/weather record, one model output and one final recommendation with timestamps. This proves the data lineage instead of only describing it.

Attach a reserve-model validation table: hold-out location, predicted grade, actual grade, absolute error and uncertainty interval.

Attach a historical shortfall replay: event date, available signals, predicted risk lead time, actual production impact and whether the recommended intervention would have been feasible.